

JEE MAIN REVISION MASTER: INDEFINITE INTEGRATION

Subject: Mathematics | Target Scope: JEE Main Previous Years | Empirical Base: 28 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Integration by Substitution & Standard Integrals	28 PYQs	Integration Methods
Standard	Integration by Parts & Partial Fractions	0 PYQs	Advanced Integration

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Integration by Substitution & Standard Integrals	$\int \frac{f'(x)}{f(x)} dx = \ln f(x) + C$, $\int \frac{dx}{x^2+a^2} = \frac{1}{a} \tan^{-1} \frac{x}{a} + C$
Integration by Parts & Partial Fractions	$\int u v dx = u \int v dx - \int \left(\frac{du}{dx} \int v dx \right) dx$, $\int e^x [f(x) + f'(x)] dx = e^x f(x) + C$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Integration by Substitution & Standard Integrals (Very High Yield — 28 PYQs)

Core Principle: Method of substitution: $\int f(g(x))g'(x)dx = \int f(u)du$. Standard algebraic and trigonometric forms: $\int \frac{dx}{x^2+a^2} = \frac{1}{a} \tan^{-1}(x/a)$, $\int \frac{dx}{\sqrt{a^2-x^2}} = \sin^{-1}(x/a)$, $\int \frac{f'(x)}{f(x)} dx = \ln|f(x)|$.

■ **Caution / JEE Pitfall:** Forgetting the factor $1/a$ in $\frac{1}{a} \tan^{-1}(x/a)$, or adding $1/a$ to $\sin^{-1}(x/a)$.

■ **Speed Trick:** For $\int e^x [f(x) + f'(x)] dx$, the answer is immediately $e^x f(x) + C$.

Representative Source PYQs:

- [2026 Shift] Q#1 (MCQ): " $\int e^{-x} \log 2 \cdot 2^x dx = \dots$ " [Ans: B]
- [2026 Shift] Q#2 (MCQ): "If 'n' is a natural number, then $\int \frac{\sin^n x \cos^{n+2} x}{\cos^2 x} dx = \dots$ " [Ans: D]
- [2025 Shift] Q#3 (MCQ): "The value of $\int \frac{dx}{(x+1)(x+2)}$ is..." [Ans: N/A]

3.2 Integration by Parts & Partial Fractions (Standard Yield — 0 PYQs)

Core Principle: Integration by parts $\int u v \, dx = u \int v \, dx - \int (u' \int v \, dx) \, dx$ guided by the ILATE priority rule. Resolution into partial fractions for proper rational functions.

■ **Caution / JEE Pitfall:** Applying partial fractions to improper rational functions without dividing numerator by denominator first.

■ **Speed Trick:** In KCET, differentiate the four given options if finding the integral takes more than 1 minute.

Representative Source PYQs:

- [2026 Shift] Q#1 (MCQ): " $\int e^{-x} \log 2 \cdot 2^x dx = \dots$ " [Ans: B]
- [2026 Shift] Q#2 (MCQ): "If 'n' is a natural number, then $\int \frac{\sin^n x \cos^{n+2} x}{\cos^n x} dx = \dots$ " [Ans: D]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2026 (Q#1)

$$\int e^{-x} \log 2 \cdot 2^x dx =$$

- (A) $\log x + C$
- (B) $x + C$ [Correct]
- (C) $(1/x) + C$
- (D) $\frac{1}{x^2} + C$

Correct Answer: B

Solution Analysis: Since 2^x can be written as $2^x = e^{x \log 2}$ the integrand becomes $e^{-x} \log 2 \cdot 2^x = e^{-x} \log 2 \cdot e^{x \log 2}$. Using $e^a \cdot e^b = e^{a+b}$, $e^{-x} \log 2 + x \log 2 = e^0 = 1$. So the integral is $\int 1 dx = x + C$. So the correct option is $x + C$ Option B.